INCH-POUND

MIL-S-20708/19G
13 March 1989

SUPERSEDING
MIL-S-20708/19F
30 July 1987

MILITARY SPECIFICATION SHEET

SYNCHRO, TORQUE RECEIVER TRANSMITTER, TYPE 15TRX4A

This specification is approved for use by all Departments and Agencies of the Department of Defense. The requirements for acquiring the Synchros described herein shall consist of this specification and the latest issue of MIL-S-20708

TABLE I. Requirements

Requirement	Value	Unit	Tolerance
Frequency	400	Hz	±1%
Primary Voltage	115	volts	±1%
Primary Current	190.0	milliamps	maximum
Primary Power	2.9	watts	maximum
Impedance			
Zss	50.0-82.0	ohms	min-max
Impedance Angle			
Zss	33.0-46.0	degrees	min-max
Transformation Ratio	0.783	-----	±2%
Phase Shift (Lead)	5.0	degrees	maximum
Electrical Error	6.0	minutes	maximum
Receiver Error	45.0	minutes	maximum
Null Voltage			
Total	100.0	millivolts	maximum
Fundamental	50.0	millivolts	maximum
Synchronizing Time			
30° ±2°	1.0	seconds	maximum
Synchronizing Time			
177° ±2°	2.0	seconds	maximum
Torque Gradient	0.030	ounce-inches per degree	minimum
Radial Play	0.0006	inches	maximum
End Play	0.0020	inches	±0.0010
Temperature Rise	35.0	degrees (C)	maximum
Variation of Voltage (+10%) and Freq. (-5%)	4.1	watts	maximum

Ⓒ denotes changes

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TABLE II Military part number variant characteristics

Military Part No 1/	A ± 010	B ± 010	G Maximum	Shaft Style	Terminal End	U 2/ + 0000 − 0002
M20708/19-01A	555	500	1 640	Splined	Terminal Board	1872
M20708/19-02A	↓	↓	1 640	Plain & Slotted	Terminal Board	↓
M20708/19-03A	↓	↓	1 378	Plain & Slotted	Wire Leads	↓

1/Military part number M20708/19-01A shall be the NATO standard

1/The suffix letter following the numerical dash number corresponds to the latest modification letter in the type designation

2/The tolerance for plain and slotted shaft diameters is + 0000 and − 0005

TABLE III Military part number cross-references

SUPERSEDED MILITARY PART NUMBERS		NEW MILITARY PART NUMBERS
MIL-S-20708/19E	MIL-S-20708/19D	
M20708/19E-001 M20708/19E-002 M20708/19E-003	M20708/19-001	M20708/19-01A M20708/19-02A M20708/19-03A

NOTE All line items shown in Table III refer to equivalent and interchangeable synchros of the same type designation modification. Part number changes do not affect form, fit, or function of the synchros listed therein.

Custodians
Army-AR
Navy-AS
Air Force-85

Review Activities
Army-MI,AV
Navy-OS
Air Force-11,17,99
DLA-ES

User Activities
Navy-MC,CG

Preparing Activity
Navy-AS

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